

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration: 1 Hour
Total Marks:50

Annexure A

Scenario Based Assignment

Code of Conduct:

I Tharish K (192412004) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Tharish*

20+20+22+20+14
796
SP

1. Autonomous Underwater Vehicle (AUV) Rescue Mission

1. Greedy Best-First Search (GBFS) Behaviour

Greedy Best-First search selects that appears closest to the goal using only the heuristic function.

$$f(n) = h(n)$$

Where:

* $h(n)$ = Estimated straight-line distance to the survivor.

In this scenario:

- * The AUV always moves toward the estimated survivor location.
- * It ignores travel cost caused by underwater currents and depths.
- * It reacts quickly because it only considers the heuristic

Strengths:

- * Very fast.
- * Uses less memory.
- * Suitable when rapid decisions are required.
- * Works well if the heuristic is accurate.

Weakness:

- * Does not guarantee the shortest or safest route.
- * May enter dangerous region with strong currents.
- * Can be misled by inaccurate sensor information.
- * May need frequent re-planning in changing environments.

Partial Knowledge	Poor	Poor	Excellent
Real-time Decision	No	Limited	Yes

v. Recommended Approach

Recommended: online search Agent (with heuristic guidance such as online A* or D* Lite)*

Justification

Requirement

online Search

Unknown map

✓

Dynamic Environment

✓

Real Time Decision

✓

Adaptive Planning

✓

Safe Navigation

✓

The drone can:

* continuously update its map.

* avoid hazards.

* adapt to new obstacles

* safely locate survivors.

Therefore, an online search approach is the most suitable because it provides adaptability, real-time decision-making and safety in uncertain environments.

2 Recommended Algorithm

Recommended : A^*

Justification

Factor	Greedy	A^*
speed	High	Moderate
optimal path	No	Yes
Safety	Low	High
Handles cost	No	Yes
Dynamic Environment	Moderate	Good

Since rescue mission require:

- * Minimum delay.
- * ~~safe~~ route,
- * reliable navigation

A^* is the best choice because it balance speed, optimality, and

1- cloud Load Balancing optimization

optimization Problem:

objective:

Minimize.

* Energy consumption

* Response Time

* Hardware Wear

Subject to:

* Thousands of servers

* Dynamic workloads

* continuous task redistribution

The objective function can be written as:

$$\text{Minimize } F = \text{Energy} + \text{Latency} + \text{wear}$$

Why Traditional Search is Inefficient

Traditional search methods:

* Explore many possible states.

* Number of possible server-task assignments is extremely large.

* Real-time requests constantly change.

Therefore:

* Search Space becomes enormous

* slow response.

* High computational cost

Hence optimization algorithms are preferred.

2 Hill climbing

Hill climbing starts with one workload distribution.

It moves to a better distribution with better performance.

i) A Search*

A* search both the actual travel cost and heuristic

$$f(n) = g(n) + h(n)$$

Where:

* $g(n)$ = Actual travel cost (depth + weight strengths + distance)

* $h(n)$ = Estimated distance to survivor

Instead of simply choosing the closest node, A* evaluates both:

- * Cost already spent
- * Estimated remaining cost

Thus, it avoids dangerous or expensive routes even if they appear!

Advantages:

- * Finds the optimal path (if heuristic is admissible).
- * Considers changing movements costs.
- * Produces safe navigation

Disadvantages:

- * Requires more computation
- * Uses more memory.
- * Slightly slower than Greedy Search.

ii) Impact of Heuristic Accuracy

Accurate Heuristic

- * Greedy Search quickly reaches the survivor
- * A* explores many nodes

- * Faster execution
- * Lower computation

Inaccurate Heuristic

- * A Greedy Search may choose wrong directions repeatedly
- * A* still finds the optimal path but explores more nodes.
- * Execution time increases.

Conclusion:

Greedy depends heavily on heuristic accuracy, while A* is more robust.

ii) Effect on Dynamic changes

Dynamic changes include :

- * Shifting debris
- * Changing underwater currents
- * Poor visibility
- * Newly blocked paths

Effect on Greedy Search:

- * Easily trapped in blocked paths
- * Requires repeated replanning
- * Performance decreases

Effect on A*

- * updates path using new costs
- * Better handles blocked routes.

Dynamic Environment

Environment change continuously.

Examples:

- * Falling debris.
- * New gas leaks.
- * structural collapse

The drone must constantly adapt.

iii/

Internal Model

The drone gradually builds a map.

Process:

1. Sense nearby environment
2. Detect hazards.
3. update internal map.
4. Mark safe and unsafe areas
5. Replan path.
6. continue exploration

This internal model improves over time.

iv/

Comparison

Feature	Uniformed search	Informed search	online
Map needed	Yes	Yes	No
Uses Heuristic	No	Yes	Co
Dynamic Environment	Poor	Moderate	

Ridge:

The optimal direction requires several simultaneously changes.

Example:

Moving one task alone does not help, but moving three tasks together greatly improves performance.

Hill climbing cannot easily follow ridges.

iii) Simulated Annealing and Genetic Algorithm

Simulated Annealing

Sometimes accept worse solutions.

Advantages:

- * Escapes local minima
- * Explores larger search space
- * Gradually converges.

Genetic Algorithm:

Inspired by biological evolution

Steps:

1. Generate population
2. Evaluate fitness
3. Selection
4. Crossover
5. Mutation
6. Repeat.

Examples:

- * Move a task from an overloaded server to an idle server
- * If performance improves, accept the move.

Repeat until no better neighboring solution exists.

Advantages:

- * Very fast.
- * Easy implementation.
- * Low memory.

Limitation:

- * Can become trapped in local optimum.
- * Cannot guarantee global optimum.
- * Sensitive to starting solution.

ii) Local Maxima, Plateaus, and Ridges

Local Minimum:

A solution better than nearby solutions but not globally best.

Examples:

Load appears balanced, but another better distribution exists elsewhere.

Plateaus:

Many neighbour states have the same evaluation

Examples:

Several task re-allocations produce identical response time
The algorithm cannot determine which direction is better.

Advantages

- * search many solutions simultaneously
- * Avoids local optimum.
- * suitable for massive cloud systems

✓ Recommended Algorithm

Recommended Genetic Algorithm

Justification

Feature	Hill climbing	Simulated Annealing	Genetic Algorithm
Local optimum	Poor	Good	Excellent
Stability	Medium	Good	Excellent
Dynamic changes	Moderate	Good	Excellent
Large search space	Poor	Good	Excellent

Because cloud environments involve:

- * thousands of servers,
- * dynamic workloads,
- * Multiple objectives.

Genetic Algorithm is the most scalable and adaptable solution

Search - and - Rescue Drone

why online search Agent?

offline search assumes:

- * complete map.
- * static environment.
- * All information available before starting

This scenario has:

- * unknown building structure.
- * Blocked passages.
- * Gas leaks.
- * New hazards discovered during exploration.

Therefore the drone must:

- * Sense.
- * Decide.
- * Move.
- * update knowledge continuously

Hence online Search is required.

(i) challenges

Partial observability

The drone cannot see the whole building.

only nearby surroundings are visible.

Problems:

- * Hidden obstacles
- * unknown previous locations
- * Incomplete information